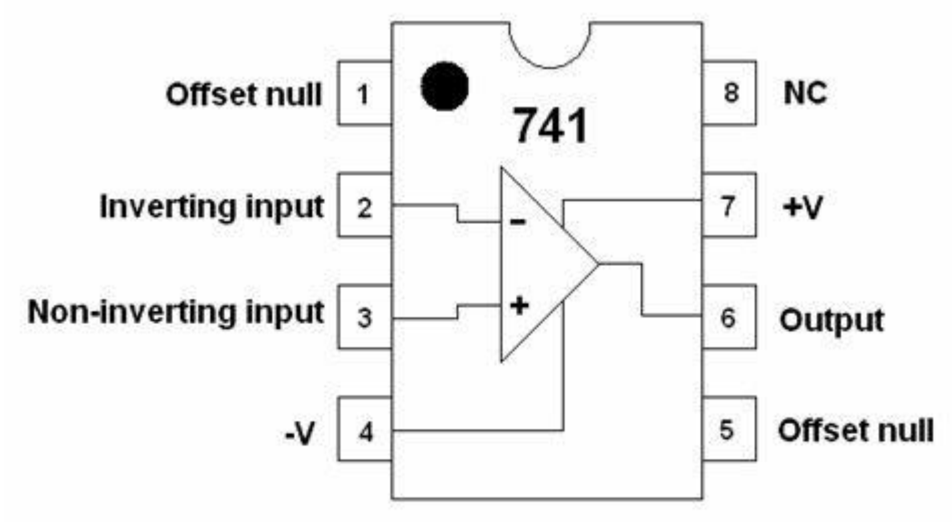
AE – (II) U1 previous year questions

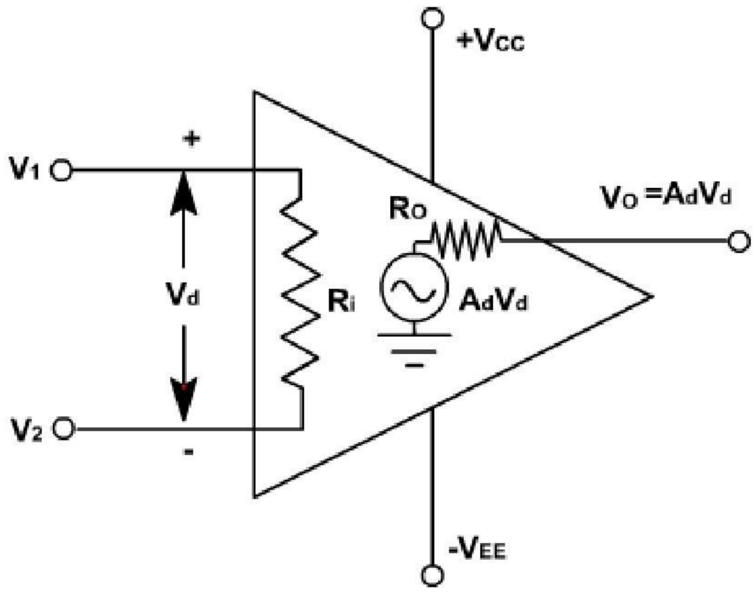
QUES1) Draw and level PIN diagram of IC-741

ANS:



QUES2) Draw and level Equivalent circuit of Op-Amp

ANS:



QUES 3) Op-amp transfer characteristicS

ANS: 1. Infinite resistance

2. zero output impedance

3. infinite open loop gain
4. infinite commom mode rejection ratio
5. infinite bandwidth
6. High input resistance
7. lower output resistance

QUES 4) EXPLAIN CMRR AND VIRTUAL GROUND

ANS:

CMRR- The CMRR in an operational amplifier is a common mode rejection ratio. Generally, the op amp as two input terminals which are positive and negative terminals and the two inputs are applied at the same point. This will give the opposite polarity signals at the output. Hence the positive and the negative voltage of the terminals will cancel out and it will give the resultant output voltage. The ideal op amp will have the infinite CMRR and with the finite differential gain and zero common mode gain.

VIRTUAL GROUND - A virtual short-circuit (or simply virtual short) refers to a condition of a differential input amplifier such as an op-amp in which its noninverting and inverting inputs have almost the same voltage. This condition is called a virtual short-circuit because the differential inputs have the same voltage even though they are not connected together. This condition is met when a negative-feedback circuit is formed using a differential amplifier with a high open-loop gain.
When the input terminal on one side is grounded to GND as shown in the figure, it is sometimes called virtual ground.

QUES 5) What is slew rate?

ANS: lew rate is defined as the maximum rate of change of an op amps output voltage, and is given in units of volts per microsecond. Slew rate is measured by applying a large signal step, such as one volt, to the input of the op amp, and measuring the rate of change from 10% to 90% of the output signal's amplitude.

QUES 6) Sketch the input and output waveform of an ideal Op-Amp in Open loop configuration for a input sign wave of I volt peak to peak.

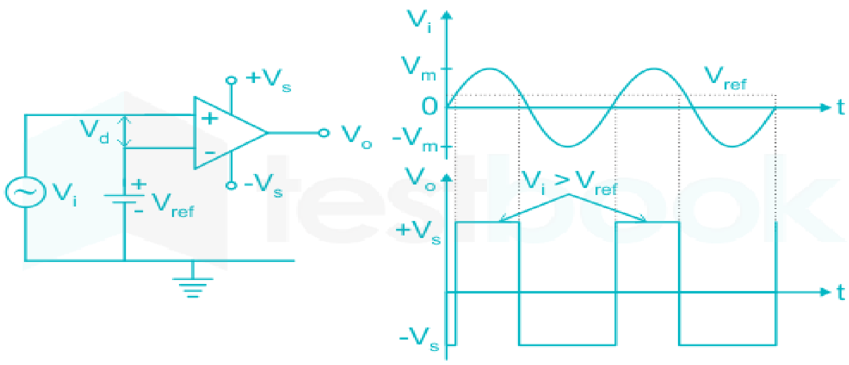
ANS:

The ideal OP-AMP circuit in the question is of a comparator:

1) If the input at the inverting terminal is more than the input at the non-inverting terminal then the output = $-V_{cc}$

2) If the input at the inverting terminal is less than the input at the non-inverting terminal then the output = $+V_{cc}$

Example:



QUES 7) What is input offset current?

ANS: The input offset current, IOS, is the difference between IB– and IB+, or IOS = IB+ - IB–. Note also that IOS is only meaningful where the two individual bias currents are fundamentally reasonably well-matched, to begin with. This is true for most voltage feedback (VFB) op amps.

QUES 8) What is input bias current and how can it be compensated? DERIVE the necessary equation to determine the value of R(comp).

ANS: he input bias current is the DC flowing into or out of an op-amp's differential inputs during operation. Designers can add resistors to neutralize the input bias current; some packages have this functionality built-in.

QUES 9) What do you mean by open loop and closed loop operation of an op-amp?

ANS: o achieve stable operation, op-amps are used with negative feedback. The gain of the device alone is called open loop gain, and the gain when configuring a negative feedback circuit is called closed loop gain. Closed-loop gain is not device-specific and is usually determined by the feedback network.

QUES 10) What are the causes of slew rate?

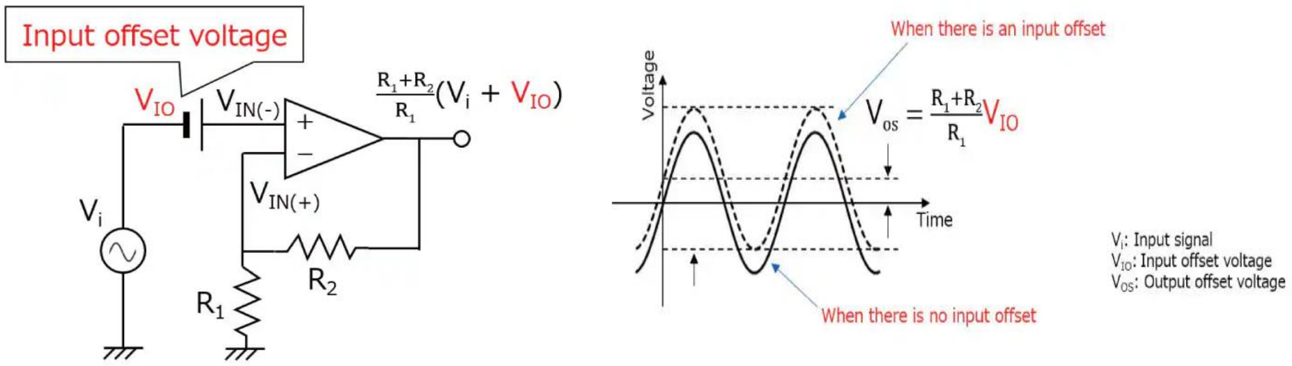
ANS: The slew rate is caused due to limited charging rate of the compensation capacitor and current limiting and saturation of the internal stages of op-amp, when a high frequency large amplitude signal is applied. For large charging rate, the capacitor should be small or the current should be large.

QUES 11) Explan the input offset voltage for inverting and non inverting amplitier with suitable diagram?

ANS: The input offset voltage (VOS) is defined as the voltage that must be applied between the two input terminals of the op amp to obtain zero volts at the output.

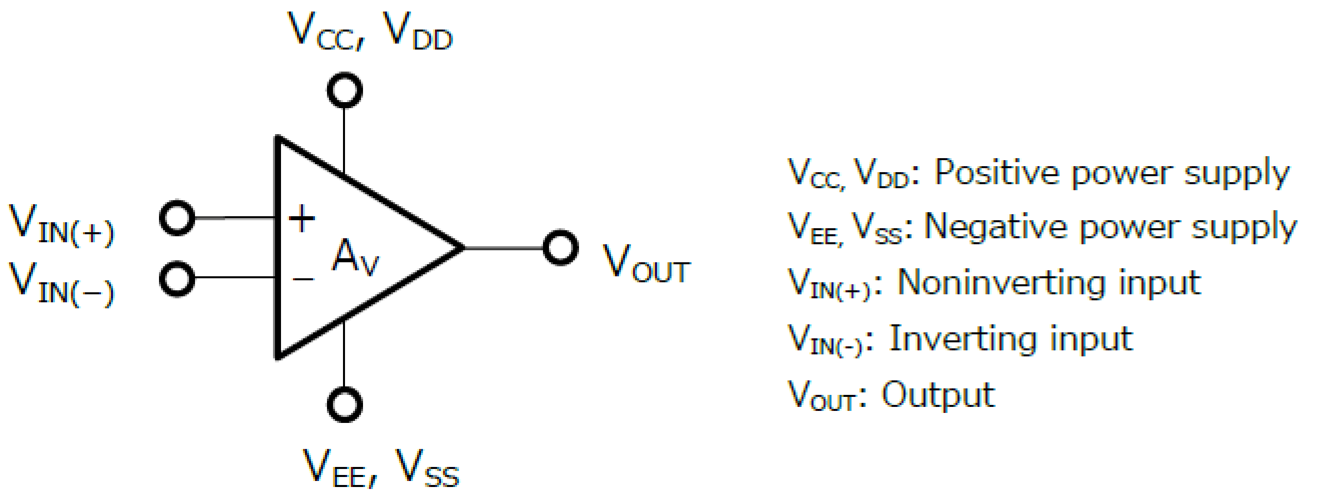
The Non inverting Op Amp

There is no input offset voltage because $V_{OS} = V_E = 0$, hence the negative input must be at the same voltage as the positive input. The op amp output drives current into R_F until the negative input is at the voltage, V_{IN}.



QUES 12) Describe the working of an op-amp with the help of its block diagram.

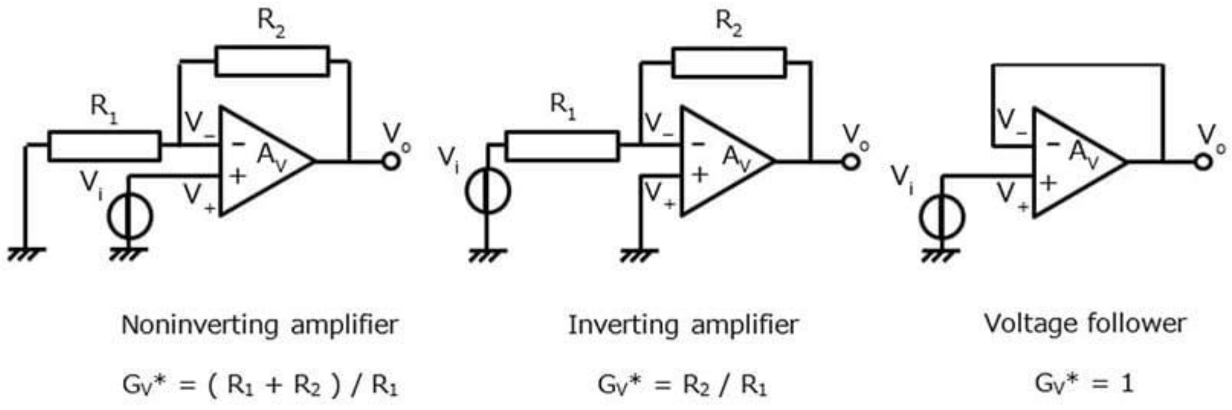
ANS:
An operational amplifier (op-amp) is an integrated circuit (IC) that amplifies the difference in voltage between two inputs.



It is so named because it was developed for perform arithmetic operations. Amplifiers, buffers, comparators, filters, etc. can be implemented with simple external circuits.
An op-amp has five terminals: positive power supply, negative power supply (GND), noninverting input, inverting input, and output. Generally, these terminals are named as shown below. (Positive and GND terminals may be omitted from the symbol of single-supply op-amps.)

An op-amp amplifies the difference in voltage between the noninverting (IN(+)) and inverting (IN(-)) inputs. Its output voltage is given by Equation 1, which indicates that the output is in the same phase as $V_{IN(+)}$ and in opposite phase to $V_{IN(-)}$.

$V_{OUT} = A * (V_{IN(+)} - V_{IN(-)})$
In the basic form of usage, an op-amp acts as a voltage amplifier or a comparator. It can also be configured as a filter, phase shifter, buffer (voltage follower), etc. Nowadays, op-amps are commonly used to amplify weak analog signals from sensors in a wide range of IoT devices and home appliances.
Op-amps are generally used with negative feedback to reduce product variations in gain and expand the bandwidth. Typical applications of op-amps include non inverting amplifiers, inverting amplifiers, and voltage followers, which are configured as shown below:



QUES 13) Explain working of Inverting and Non-Inverting Amplifiers.

ANS:

INVERTING AMPLIFIER –

The inverting op-amp or operational amplifier is an essential op-amp circuit configuration that uses a negative feedback connection. As the name suggests, the amplifier inverts the input signal and changes it.

An inverting op-amp is a type of operational amplifier circuit used to generate an output that is out of phase as compared to its input through 180 degrees which means, if the input signal is positive (+), then the output signal will be opposite. The inverting op-amp is designed through an op-amp with two resistors.

Inverting Op-Amp Working & Operations

The voltage gain provided from the above circuit is given as

$$Av = Vo/Vi$$

Where,

$$Vi-V1 = IiRi$$
$$V1-V0 = IfRf$$

But, we know that a perfect operational amplifier includes unlimited input impedance because there is no flow of current into its input terminals. I1 = I2 = 0. Therefore, Ii is equivalent to If. So,

$$Vi-V1 = IfRi$$

$$V1-V0 = IfRf$$

We already know that in a perfect operational amplifier, the voltage at two inputs in the op-amp is always equivalent.

Because the non-inverting terminal is grounded, so the voltage that appeared at the noninverting terminal is V1 = V2= 0. So, the equation will be,

$$Vi-0 = IfRi$$
$$0-Vo = IfRf$$

From the above equations, we can get,

$$-Vo/ Vi = IfRf/ IfRi$$
$$Vo/ Vi = – (IfRf/ IfRi)$$
$$Vo/ Vi = – (Rf/Ri)$$

This equation can be changed to get Vout is

$$Vout/ Vi = – (Rf/Ri)$$
$$Vout = – (Rf/Ri) \times Vin$$

The inverting op-amp's voltage gain is,

Inverting Op Amp Gain **(Av) = – (Rf/Ri)**

these amplifiers provide outstanding linear characteristics to make them ideal. In addition, they are frequently used to change the i/p current to the o/p voltage in the Transresistance form otherwise Transimpedance Amplifiers form.

The output voltage (Vout) equation shows that the op-amp circuit is linear for a fixed gain of an amplifier like Vout = Vin x Gain. So this property is very helpful in changing a small sensor signal to a better voltage.

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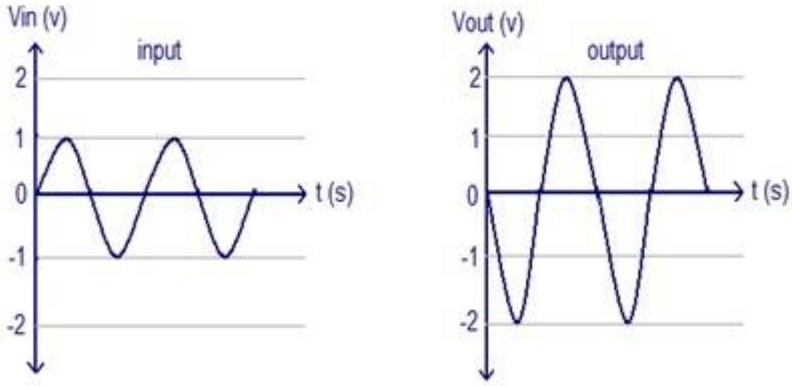
Inverting Op Amp Waveforms

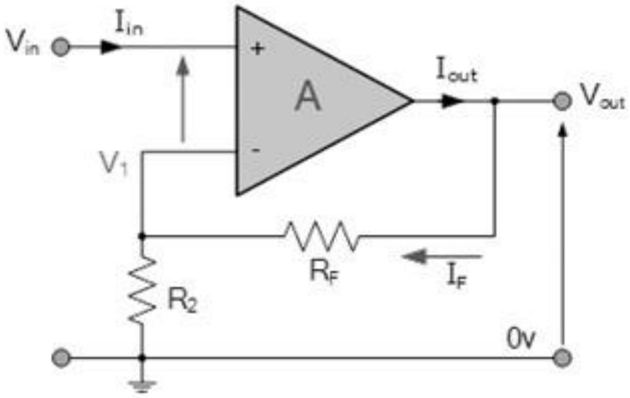
The inverting op-amps input & output waveforms are shown below. The following waveforms can be drawn by assuming the gain of the amplifier & the sine wave is an input signal. From the following waveforms, it is very clear that the output is double in magnitude as compared to the input like Vout = Av * Vin & phase is reverse to the input.



NON INVERTING AMPLIFIER -

Non-inverting op-amp definition is, when the output of an operational amplifier is in phase with an input signal then it is known as a non-inverting op-amp. In this amplifier, the input signal is applied to the +ve terminal of an operational amplifier. A non-inverting amplifier generates an amplified output signal that is in phase with the applied input signal.

The non-inverting op-amp circuit diagram is shown below. In this circuit configuration, the output voltage signal is given to the inverting terminal (-) of the operational amplifier like feedback through a resistor where another resistor is given to the ground. Here, a voltage divider with two types of resistors will provide a small fraction of the output toward the inverting pin of the operational amplifier circuit.



Non-Inverting Op-Amp Circuit

These two resistors will provide necessary feedback to the operational amplifier. In perfect condition, the op-amp's input pin will provide maximum input impedance whereas the output pin will provide low output impedance.

Non-Inverting Op-Amp Voltage Gain

The flow of current through the R1 resistor can be given as "**Vin/R1**".

Based on the current rule, both the inputs don’t draw current, thus the flow of current will be throughout R2.

After that, the output voltage (Vo) can then be **Vout = Vin + (Vin/R1)*R2**. The non inverting op-amp gain formula is **Av = Vout/Vin = 1+ (R2/R1)**. Here, the gain value should not be < 1. Therefore the non-inverting op-amp will generate an amplified signal that is in phase through the input.

In the above equation Av = Op-amp’s voltage gain

‘R2’ is a feedback resistor

‘R1’ is a resistor connected to the ground.

Input Impedance

In a non-inverting operational amplifier circuit, the input impedance (Zin) can be calculated by using the following formula.

$$Zin = (1+ A\alpha\ \beta)*Zi$$

In the above equation, ‘Aα’ is an open-loop voltage gain

‘Zi’ is the input impedance of op-amp without using feedback

‘β’ is a feedback factor

So, the feedback factor for a non-inverting amplifier can be calculated as

$$\beta = R2/(R1+R2)$$
$$\beta = 1/ ACL$$

So, for a non-inverting operational amplifier circuit, the input impedance (Zin) can be calculated as

$$Zin = ((1 + (A\alpha / ACL))*Z1$$

Output Impedance

In the non-inverting operational amplifier, the output impedance can be measured as

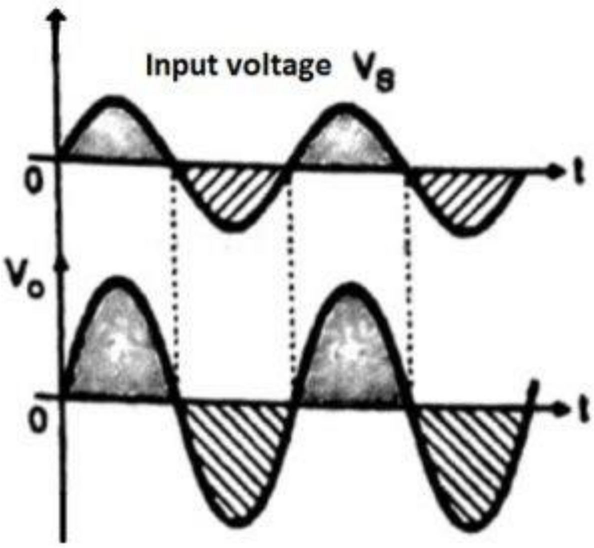
$$Zout = Zo/(1+ A\alpha\ \beta)$$

We know the feedback factor **β = 1/ACL**, so the output impedance for a non-inverting op-amp can be calculated as

$$ZOUT = Zo /(1 + (A\alpha /ACL))$$

Non Inverting Op Amp Waveform

The input & output waveforms of non-inverting op-amp waveforms are shown below.



Output Waveforms

So the positive sign specifies that there is no phase shift between input & output. The voltage gain is dependent on two resistances R1 and Rf. By changing the values of the two resistances required gain can be adjusted.

Analog Electronics – II

ECC - 214

Unit 2 PYQs

1.) Give a brief explanation on the following:

<u>CMRR</u>- Known as Common Mode Rejection Ratio.

- Common Mode is when same voltage is applied on both input terminals of the op-amp.
- CMRR is the differential Gain (Ad) to the Common Mode gain (Acm).

$$CMRR = \frac{A_{DM}}{A_{CM}}$$

$$CMRR_{dB} = 20 \log_{10} \frac{A_{DM}}{A_{CM}}$$

$$= 20 \log_{10} CMRR$$

$$A_D = \frac{V_O}{V_{IN}}$$

$$A_{CM} = \frac{V_{OCM}}{V_{CM}}$$

- CMRR ratio of 741 IC is 90db. And Ideal Op-amp will have infinite CMRR and with finite differential gain and zero common mode gain.

<u>SVRR</u>- Known as Supply Voltage Rejection Ratio.

- The change in an op-amp input offset voltage Vio caused by variations in the supply voltage is called SVRR.
- Expressed in microvolts per volts or in decibels.

$$SVRR = \frac{\Delta Vio}{\Delta V}$$

- SVRR ratio of 741 IC is 150 µV/V.

<u>Input Offset Voltage</u>- Denoted by Vio

- A small voltage applied to the input terminals to make the output voltage zero when the input terminals are grounded.
- 2 mV for 741 IC.

Input Offset Current- Denoted by I_{io}

- The difference between Input Bias currents at the input terminals of the op-amp.

$$I_{io} = I_{B1} - I_{B2}$$

- 200 nA for 741 IC.

Input Bias Current- Denoted by I_B

- Average value of the two input bias currents.

$$I_B = \frac{I_{B1} + I_{B2}}{2}$$

Where, I_{B1} = dc bias current flowing into the noninverting input
I_{B2} = dc bias current flowing into the inverting input

- 500 nA for 741 IC.

Slew Rate- Denoted by V/µs.

- Change in the response of an op-amp with change in input voltage in unit time or the maximum rate of change of output voltage.

$$SR = \frac{\Delta V_o}{\Delta t}$$

- In early stages, 741 IC had a SR of 0.5 V/µs but now 741 IC's have a SR of 70 V/µs.
- More the SR, better the IC will perform.

Thermal Drift- Values of V_{io}, I_B, and I_{io} vary with:

1. Change in temperature
2. Change in supply voltages: $+V_{CC}$ and $-V_{EE}$
3. Time

- Hence, the average rate of change of Input offset voltage, Input Bias and Input offset
current per unit change in temperature is called Thermal drift.

$\frac{\Delta I_{io}}{\Delta T}$ = thermal drift in the input offset current (pA/°C)

$\frac{\Delta I_{B}}{\Delta T}$ = thermal drift in the input bias current (pA/°C)

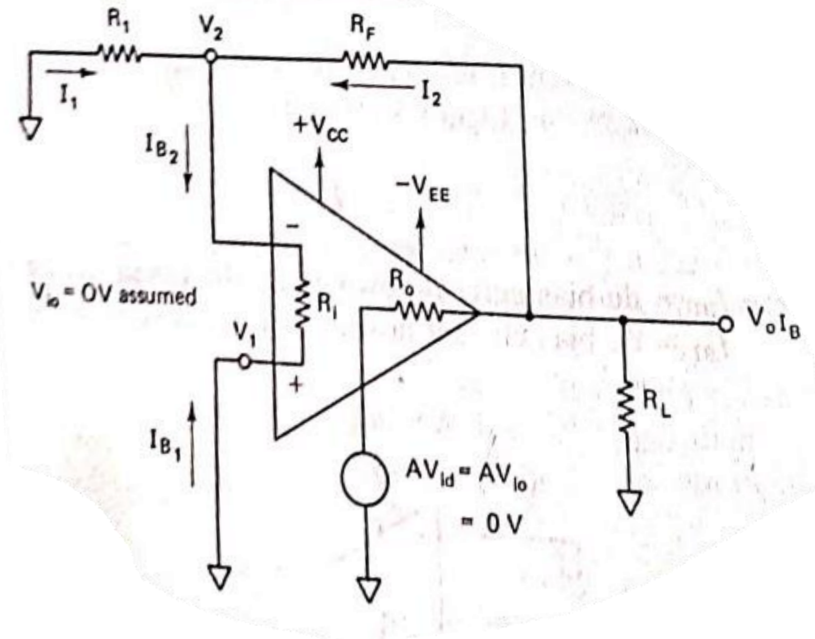
$\Delta V_{io}/\Delta T.$ = thermal drift in input offset voltage ($\mu V/°C$)

$$\frac{\Delta V_{io}}{\Delta T} = \text{thermal voltage drift} \quad 15\ \mu\text{V/°C maximum}$$

$$\frac{\Delta I_{io}}{\Delta T} = \text{thermal current drift} \quad 200\ \text{pA/°C maximum}$$

2.) How can Input bias current be compensated. Deriving necessary equations find the value of Rom.

- To compensate the effect of bias current, a Rom/Rcomp resistor is connected across the non-inverting terminal of an op-amp. The value of the compensating resistor is equal to the parallel combination of R1 and Rf.



$$V_2 = (R_1 || R_F) I_{B2}$$

$$V_2 = \frac{R_1 R_F}{R_1 + R_F} I_{B2}$$

Writing a node voltage equation for node V_2, we get

$$I_1 + I_2 = I_{B2}$$

$$\frac{0 - V_2}{R_1} + \frac{V_{oI_B} - V_2}{R_F} = \frac{V_2}{R_i}$$

where V_{oI_B} = output offset voltage due to input bias current
R_i = input resistance of the op-amp (see Figure 5-16)

Rearranging Equation (5-13), we get

$$\frac{V_{oI_B}}{R_F} = V_2 \left(\frac{1}{R_1} + \frac{1}{R_F} + \frac{1}{R_i}\right)$$

Since R_i is extremely high (ideally ∞), $1/R_i \cong 0$ siemens. Therefore,

$$\frac{V_{oI_B}}{R_F} = V_2 \frac{R_1 + R_F}{R_1 R_F}$$

Substituting in Equation (5-14) the value of V_2 from Equation (5-12), we get

$$V_{oI_B} = \frac{R_1 R_F I_{B2}}{R_1 + R_F} \left(\frac{R_1 + R_F}{R_1}\right) \tag{5-15}$$

$$V_{oI_B} = R_F I_{B2}$$

From Equation (5-10),

$$V_{oI_B} = R_F I_B \tag{5-16}$$

$$V_2 = R_p I_{B2}$$

Where,

$$R_p = \frac{R_1 R_F}{R_1 + R_F}$$

$$V_1 = R_{OM} I_{B1}$$

$$R_p I_{B2} = R_{OM} I_{B1}$$

$$R_p = R_{OM}$$

or

$$\frac{R_1 R_F}{R_1 + R_F} = R_{OM}$$

3.) A non-inverting amplifier with a gain of 100 is nulled at 25℃. What will happen to the output voltage if the temperature rises to 50℃ for an offset voltage drift of 0.15 mV/C?

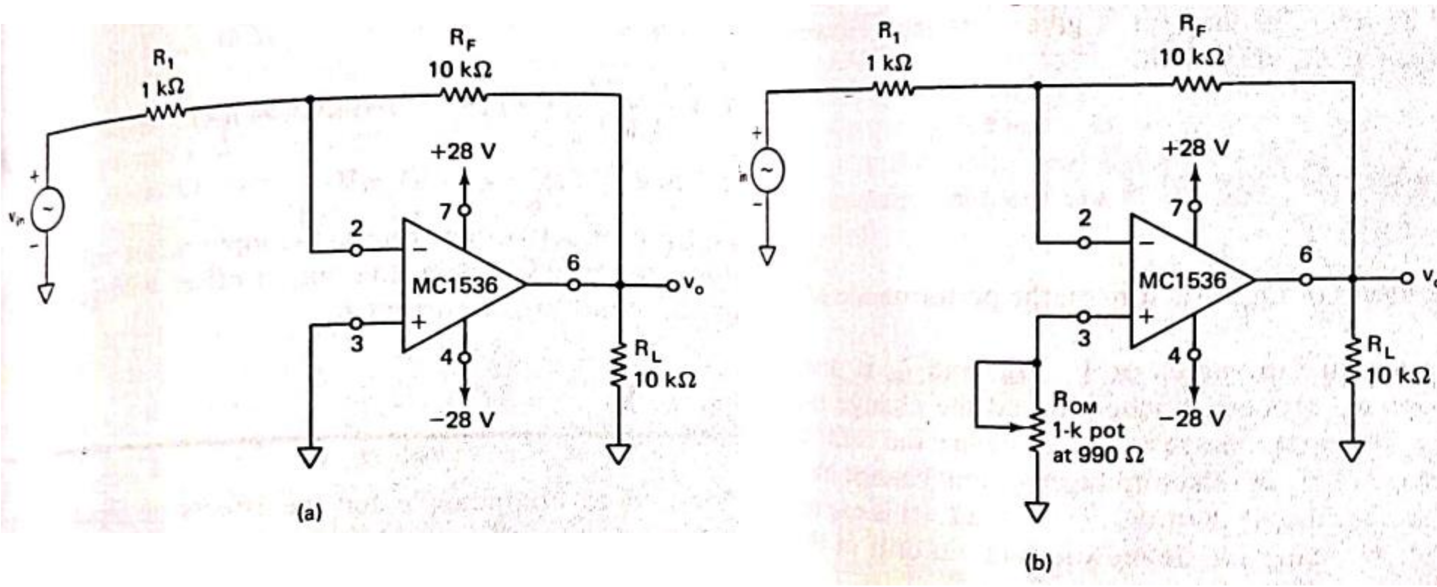
Input offset voltage due to temperature rise = 0.15 mV/°C × (50°C − 25°C) = 3.75 mV. Since this is an input change, the output voltage will change by

$$V_o = V_{os} \times A_{CL}$$

$$= 3.75 \text{ mV} \times 100 = 375 \text{ mV}$$

This could represent a very major shift in the output voltage.

4.) Compute the maximum total output offset voltage in the circuit shown below:



(a) (b)

a)

$$V_{ooT} = \left(1 + \frac{R_F}{R_1}\right) V_{io} + (R_F)I_B$$

$$= \left(1 + \frac{10 \text{ k}\Omega}{1 \text{ k}\Omega}\right)(7.5 \text{ mV}) + (10 \text{ k}\Omega)(250 \text{ nA})$$

$$= 82.5 \text{ mV} + 2.5 \text{ mV} = 85 \text{ mV}$$

b) $$V_{ooT} = \left(1 + \frac{R_F}{R_1}\right) V_{io} + (R_F) I_{io}$$

$$= \left(1 + \frac{10\ \text{k}\Omega}{1\ \text{k}\Omega}\right)(7.5\ \text{mV}) + (10\ \text{k}\Omega)(50\ \text{nA})$$

$$= 82.5\ \text{mV} + 0.5\ \text{mV} = 83\ \text{mV}$$

5.) **An op-amp having a slew rate of 62.8 V/μsec is connected in a voltage follower configuration. If the maximum amplitude of the input sinusoidal is 10V, then the minimum frequency a which the slew rate limited distortion would set in at the output is**

$$\text{Slew rate} = \frac{\Delta V_0}{\Delta V_i} \times \frac{\Delta V_i}{\Delta t} = A_{CL}\,(2\pi f_m V_m)$$

Given that, slew rate = 62.8 V/μsec

$62.8 \times 10^6 = 6.28 \times 10 \times f_m$

$f_m = 1$ MHz

6.) Given that CMRR is 100 db. Input Common mode Voltage is 12V. Differential voltage gain is 4000. Calculate output common-mode voltage.

Given: A_d = 4000,

CMRR = 100 db

⇒ CMRR in db = 20 log |CMRR| = 100

⇒ CMRR = 100000

$$100000 = \frac{4000}{A_c}$$

Input common-mode voltage is 12 V

⇒ V_C

Output common-mode voltage of Op-Amp will be:

$V_o = A_c V_c$

= .04 × 12 = .48 V

7.) What is the effect of a high frequency on slew rate of an op-amp?

- Since for frequency compensation we use internally compensated capacitors for op-amp, a high frequency can have an impact on the slew rate of an operational amplifier (op amp). The slew rate is the maximum rate of change of the output voltage and is typically specified in volts per microsecond.
- At higher frequencies, the op amp may not be able to respond as quickly to changes in the input signal, leading to a reduction in the slew rate. This can result in distortion or other undesirable effects in the output signal. It's important to consider the frequency response of the op amp when designing circuits to ensure proper performance.

8.) What are the advantages of active filters over passive filters?

- Active filters are built around op-amps or transistors. Passive filters use only inductors, capacitors, and resistors. Active filters have the following advantages over passive filters:
  (1) gains can be greater than one;
  (2) active filters can be cascaded without loading and impedance matching issues;
  (3) active filters can be constructed without using inductors, which are bulky;

(4) Unlike passive, active filters provide mechanism for gain adjustment;
(5) Passive filters are easy to construct and relatively inexpensive;

9.) Discuss the frequency response of ideal and practical filters.

i) Ideal Filters:

Ideal filters are theoretical constructs that have perfect frequency response characteristics. They can be categorized into different types based on their frequency passbands:

- **<u>Low-pass filter:</u>** Allows low-frequency signals to pass through while attenuating high-frequency signals.
- **<u>High-pass filter:</u>** Permits high-frequency signals to pass through while attenuating low-frequency signals.
- **<u>Band-pass filter:</u>** Passes a range of frequencies between two cutoff frequencies while attenuating frequencies outside this range.
- **<u>Band-stop filter (notch filter):</u>** Attenuates a specific range of frequencies while passing frequencies outside this range.

<u>Characteristics of ideal filters include</u>:

- Perfect passband attenuation and stopband attenuation.
- Infinite slope at the cutoff frequency.
- Sharp transition between the passband and stopband.
- The frequency response of an ideal filter is represented by sharp transitions between passbands and stopbands, with no ripples or distortions in the passband. However, ideal filters are purely theoretical and cannot be realized in practice due to limitations such as component tolerances, finite bandwidth, and causality.

ii) Practical Filters:

Practical filters are designed to approximate the behavior of ideal filters while considering real-world constraints.
These constraints include component tolerances, finite precision in signal processing, and trade-offs between different design parameters.

<u>Characteristics of practical filters include:</u>

- Finite passband attenuation and stopband attenuation.

- Transition bands with finite width, resulting in ripples in the frequency response.
- Limited slope at the cutoff frequency compared to ideal filters.
- Practical filters can be implemented using various analog or digital techniques, such as passive RC filters, active filters, digital finite impulse response (FIR) filters, or digital infinite impulse response (IIR) filters. Each implementation has its advantages and limitations, affecting the frequency response characteristics.

10.) Derive expression of open loop voltage gain of an op-amp as function of frequency.

Let us now obtain an expression for the gain as a function of frequency. From Figure 5–2, using the voltage-divider rule, we get

$$v_o = \frac{-jX_C}{R_o - jX_C}(Av_{id})$$

Since $-j = 1/j$ and $X_C = 1/2\pi fC$,

$$v_o = \frac{1/j2\pi fC}{R_o + 1/j2\pi fC}(Av_{id})$$

$$= \frac{Av_{id}}{1 + j2\pi fR_oC}$$

Hence the open-loop voltage gain is

$$A_{OL}(f) = \frac{(v_o)}{(v_{id})}$$

$$A_{OL}(f) = \frac{A}{1 + j2\pi fR_oC}$$

Let $f_o = 1/2\pi R_oC$; then

$$A_{OL}(f) = \frac{A}{1 + j(f/f_o)} \tag{5–1}$$

where $A_{OL}(f)$ = open-loop voltage gain as a function of frequency
A = gain of the op-amp at 0 Hz (dc)
f = operating frequency (Hz)
f_o = break frequency of the op-amp (Hz)

ANALOG ELECTONICS-2

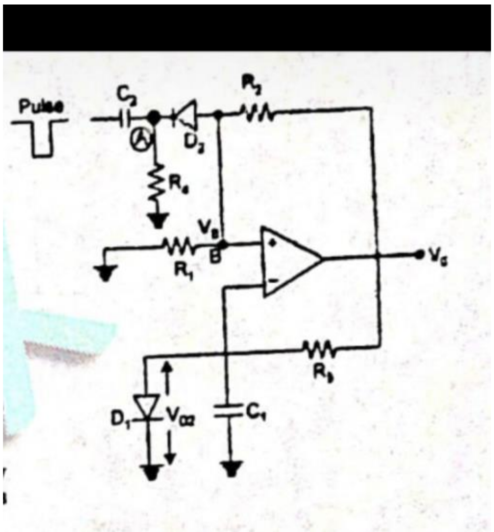
UNIT-4

Qn1) Explain Monostable Multivibrator.

Ans) Monostable multivibrator: It has only one stable state (V = V0)

Case I: If triggering pulse is not connected

D1 will conduct through R3 and VC1 = VD1=0.7 volt



For R >> R_1, diode D, will be conducting very small current

Negative triggering

Now Vb=v0*R1/R1+R2

Vb=BV0

Because Vb >>Vc means the stable state is maintained.

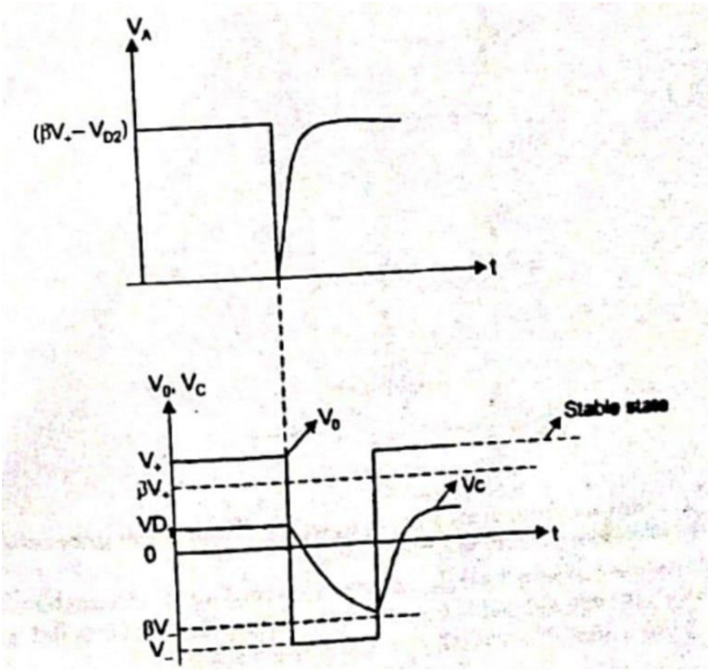
\Therefore v0=V+

CASE-2

If negative triggering pulse is connected D2 will be conducting very heavily and C2 changes towards Bv+ -BD2

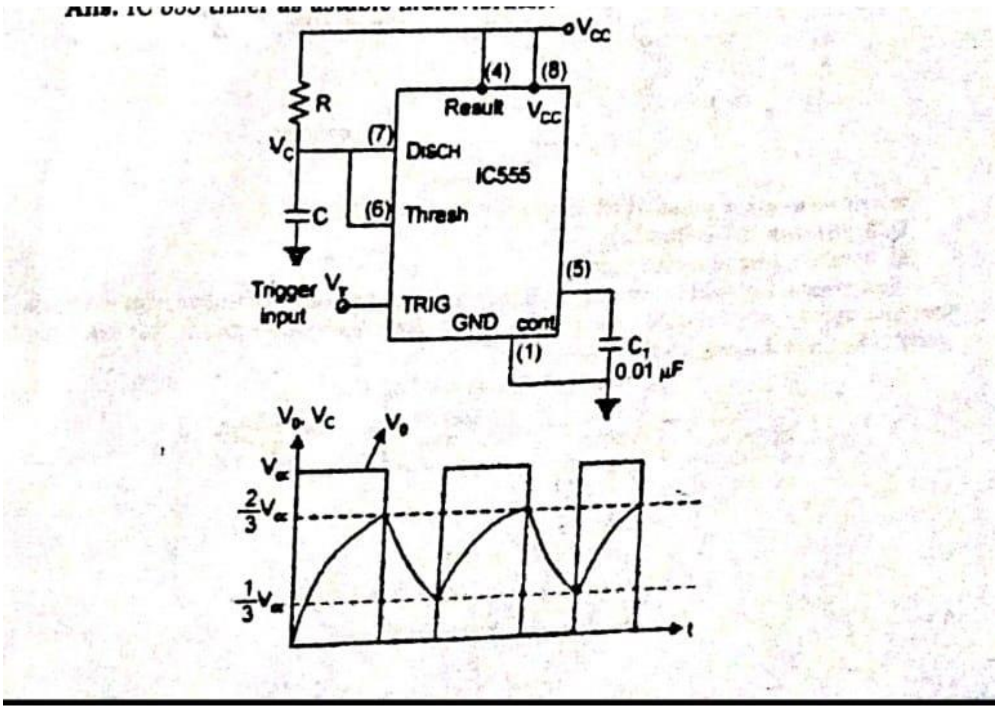
Now, Vb < vc and Op-amp switching to V- Therefore V0=V-

Now diode D1 will be reverse biased and C1 being discharged exponentially towards V- at Vc<Vb Op amp switches back to V+ wave forms.



Qn2) Discuss applications of IC 555 timer as astable and monostable multivibrator.

Ans) IC 555 timer as astable multivibrator.



Case 1: If V0=1 (High)

→Capacitor starts charging for V trough R1 and R2, Case II:

If V0> 1/3Vcc

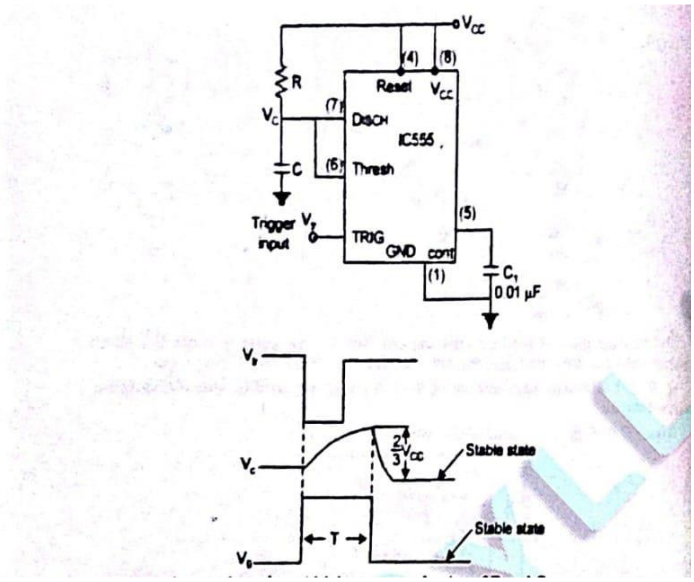
No Change and V0< 2/3Vcc (capacitor will continue to charging).

Case III: If V0 >2/3 Vcc

Now capacitor starts discharging through R,

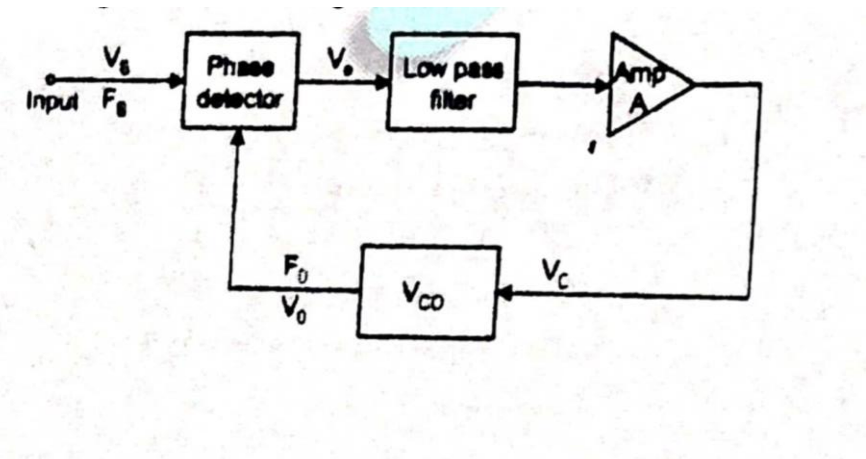
IC 555 timer as monostable multivibrator:

A negative going input trigger pulse produces a single output pulse wit predetermined width. The width of the output pulse is determined by t=1Rc.



Qn3) Explain the IC phase locked loop.

Ans) Phase Locked Loop: It is an important building block of linear system which has wide applications in radar communication and satellite communication. It is available as IC. The block diagram is shown in figure.



PLL has following component:

1. Phase detector: It is used to compare magnitude and frequency of input signal and output of Vcr.

For this purpose, two types phase detectors are used.

(a) Analog phase detector

(b) Digital phase detector

In analog phase detector transistor is used for detection and in digitial phase detector

exclusive or gate is used.

2. Low Pass Filter: It filter high frequency signals like (f, +/ signals.

3. Amplifier. It amplifies the error signal.

4. VCO. Voltage controlled oscillator is main component of PLL. It generates voltage and frequency corresponding to error voltage amplified by error amplifier.

Qn4) Explain the IC voltage regulator.

Ans. A voltage regulator provides de output voltage that is practically independent of the input voltage, output load current and temperature. The voltage regulator is one part of a power supply.

Most voltage regulators fall into two broad categories -Linear regulators and Switching Regulators Linear series regulators Linear regulator - Linear shunt regulators. The most popular of regular are the tree-

terminal fixed voltage regulator and adjustable voltage regulator. Switching regulators one also widely used.

1.78XX Seven output voltage option are there →XX indicate output voltage.

2.79XX fixed output → Negative voltage regulator

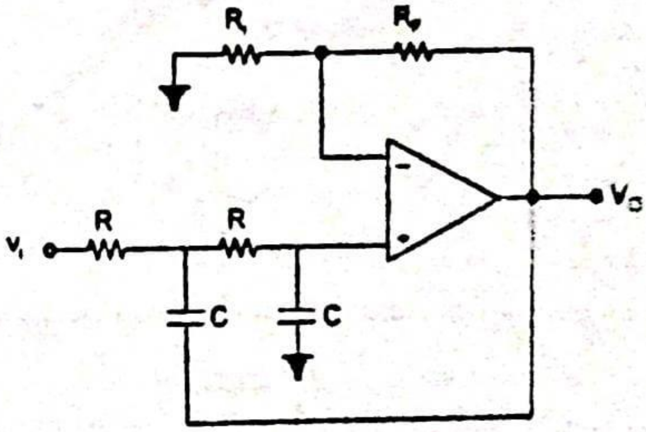
→Two extra voltage of-20 and-5.2V available.

3. LM 100 series, 200 series, 300 series.

Qn5) Discuss Butter Worth and Chebyshev filters.

Ans)

$$H(s) = \frac{A_0}{S^2C^2R^2 + SCR(3 - A_0) + 1} = \frac{A_0 \dfrac{1}{C^2R^2}}{S^2 + \dfrac{S(3 - A_0)}{C^2R^2} + \dfrac{1}{C^2R^2}} \quad \text{...(1)}$$



Comparing with standard transfer function

$$H(s) = \frac{A_0\, \omega_h^2}{s^2 + \alpha\, \omega_h s + \omega_h^2} \quad \text{...(2)}$$

where A_0 = Gain

ω_h = upper cut-off frequency

α = damping coefficient

$$\omega_h = \frac{1}{RC}$$

$$\alpha = (3 - A_0)$$

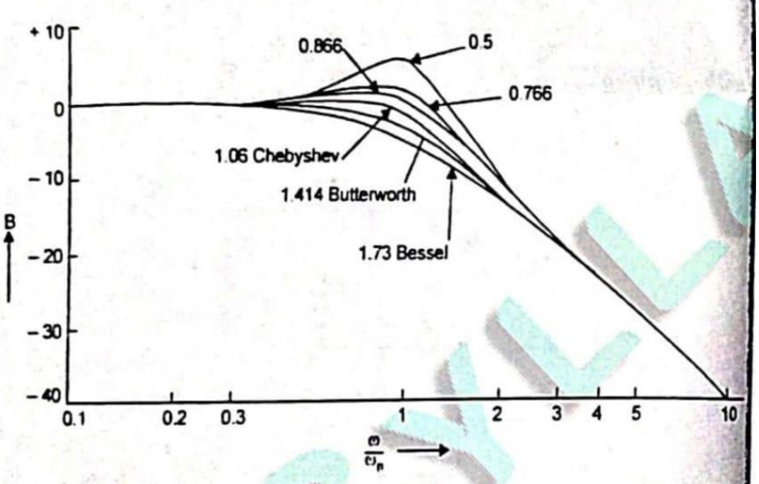
by equation (2)

$$H(s) = \frac{A_0}{\frac{s^2}{w_h^2} + \frac{s}{w_h}\alpha + 1} = \frac{A_0}{\left(\frac{jw}{w_h}\right)^2 + j\alpha\left(\frac{w}{w_h}\right) + 1}$$

The expression of magnitude in dB

$$20 \log |H(s)| = 20 \log \frac{A_0}{1 + j\alpha\left(\frac{w}{w_h}\right) + \left(\frac{jw}{w_h}\right)^2}$$

$$= 20 \log \frac{A_0}{\sqrt{\left(1 - \frac{w^2}{w_h^2}\right)^2 + \left(\alpha \frac{w}{w_h}\right)^2}}$$

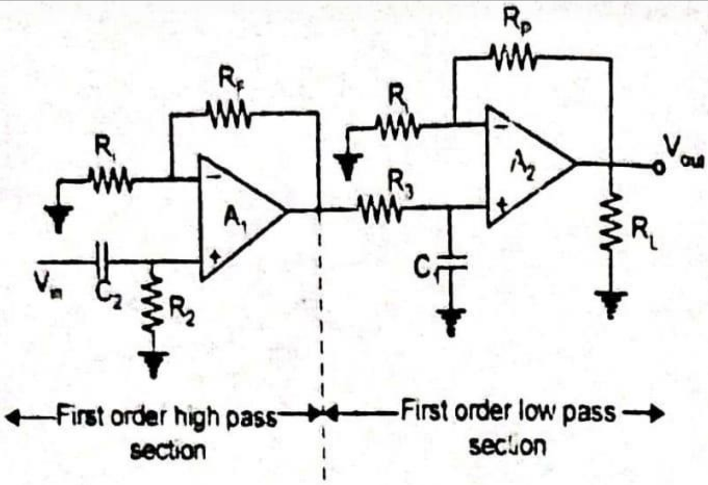


d

Fig. Second order low-pass active filter response for different damping (unit gain $A_0 = 1$)

For Butterworth filter, the damping coefficient (a) should be 1.414 to get flat pass band. Audio filters are usually butter worth.

• Chebyshev filters are more lightly damped, that is, the damping coefficient an 1.06, this increases overshoot and ringing occurs deteriorating the pulse response U advantage however, is a faster initial roll-off compared to butter worth. The frequency response for different value of a is shown in fig.



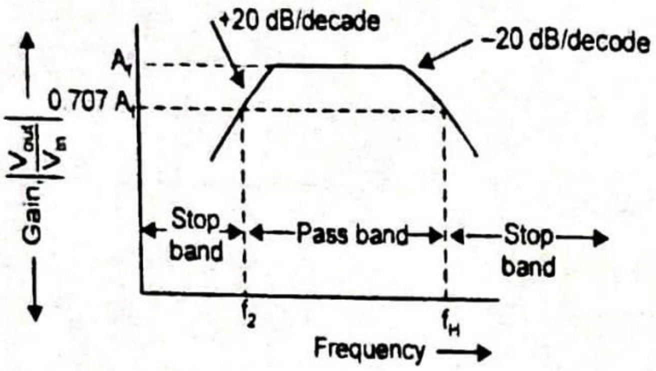


Fig. Frequency Response

$$= \frac{A_{01}\, f/f_e}{\sqrt{1+(f/f_e)^2}}$$

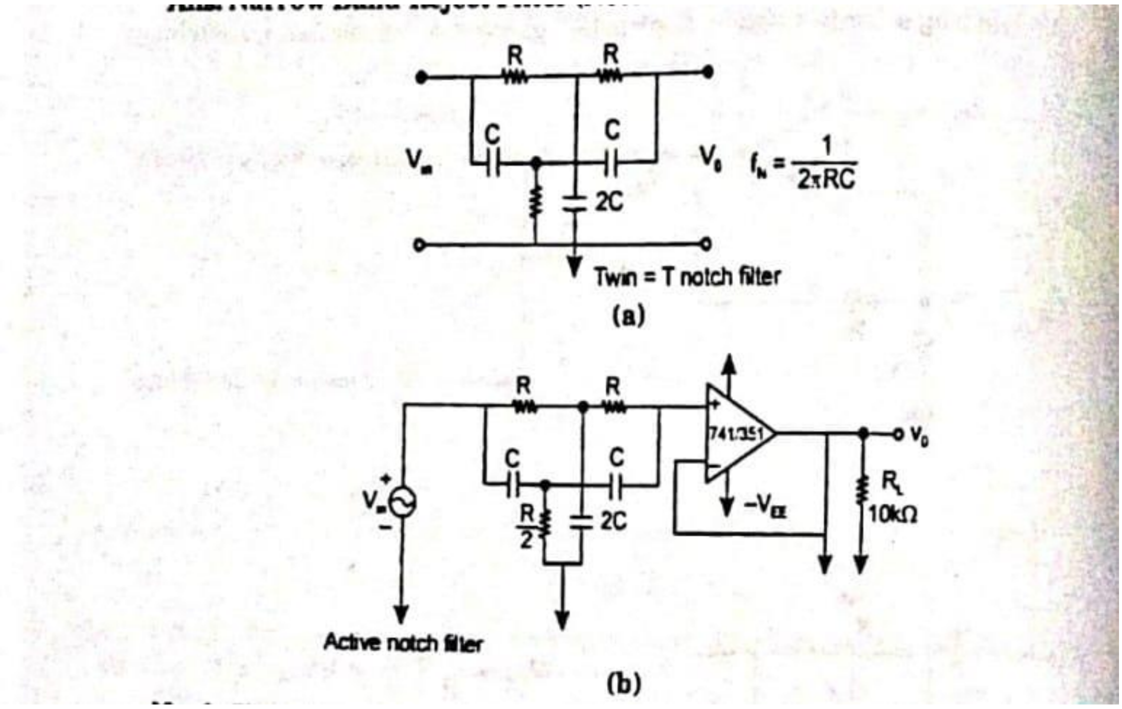
Where $$f_e = \frac{1}{2\pi R_2 C_2}$$

Similarly $$|H_{LP}| = \frac{A_{02}}{\sqrt{1+(f/f_h)^2}}$$

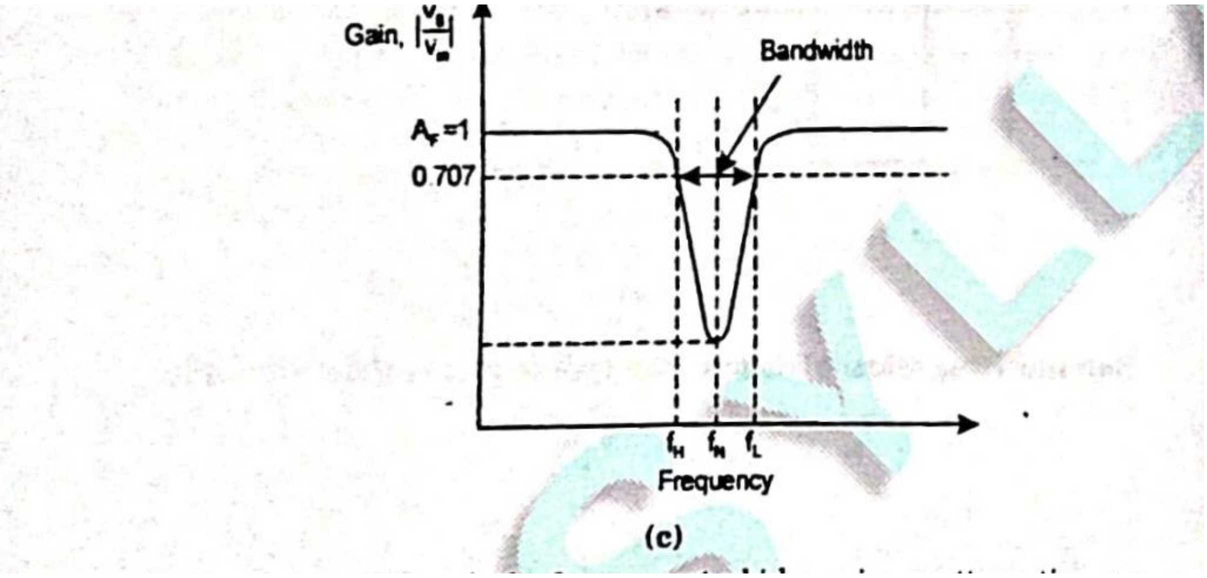
where $$f_h = \frac{1}{2\pi R_1 C_1}$$

Qn6) Discuss Notch filter and its filter function.

Ans) Notch filter -



Notch filter is used to reject a single frequency.



fN=Notch out frequency, is the frequency at which maximum attenuation occurs. The most common use of notch filter is in communications, and biomedical instruments, for eliminating undesired frequencies.

Qn7) Explain the Band Pass Filter (BPF).

Ans) Band pass filter (B.P.F):

A band pass filter passes all signal lying within a band between a lower- frequency limit and an upper-frequency limit and essentially rejects all other frequencies that are outside this specific band.

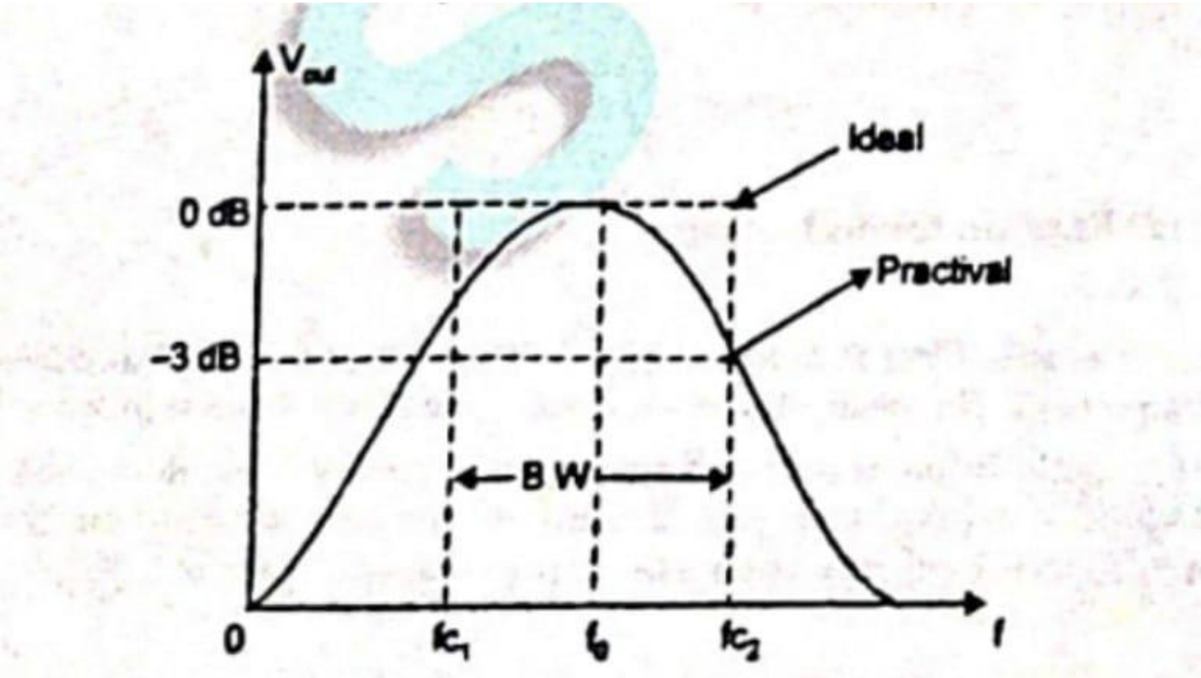


Fig. Frequency Response of B.P.F

Qn8) Explain the Low Pass filter.

Ans) (i) LPF

A low pass filter is one that passes frequencies from de (zero frequency) to ft (cut off frequency). The pass band of ideal low pass filter is equal to fc (B. W-fc The response drops to zero at frequencies beyond the passband the gain almost constant until the frequency is at the critical frequency. Aftar this the gain drops rapidly at a fixed roll-off rate as shown in the response curve of L.P.F.

1 The cut off frequency of simple LPF =>fc=1/2*3.14*RC.

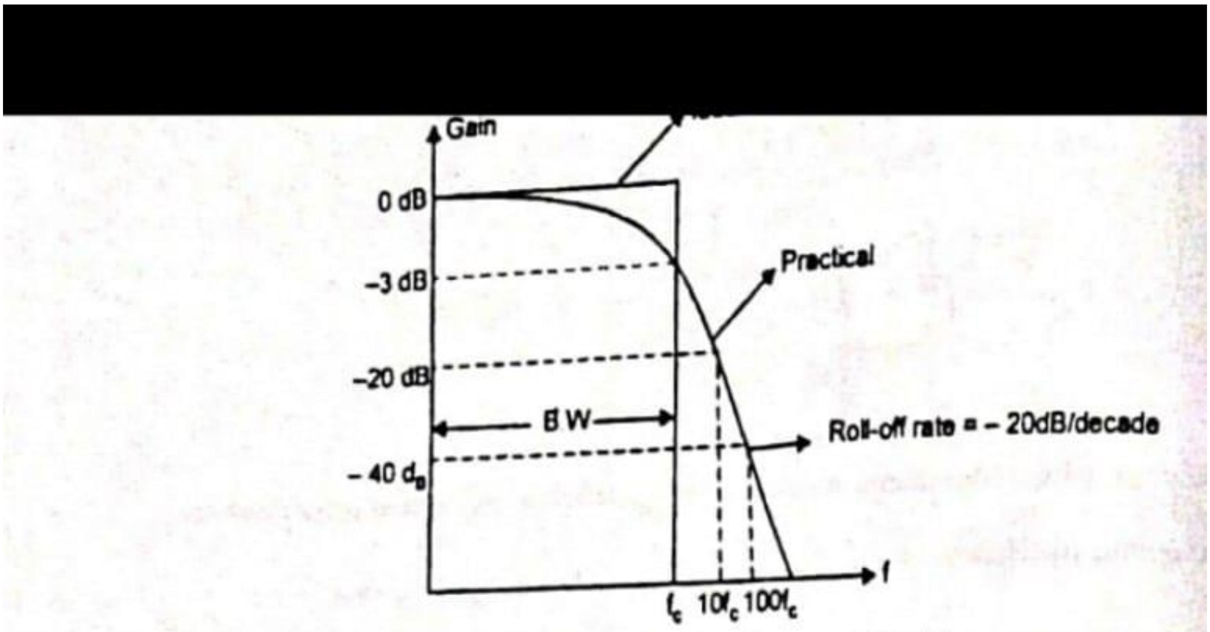


Fig. Response Curve of L.P.F.

Qn9) Explain the High Pass filter.

Ans) High pass filter (H.P.F):

A high pass filter is one that significantly attenuate or rejects all frequencies below fe and passes all frequencies above fc. The critic frequency (cut-off) is the frequency at which the output is 70.7% (-3 dB) as shown figure. The pass band of H.P.F. is all frequencies above the critical frequency.

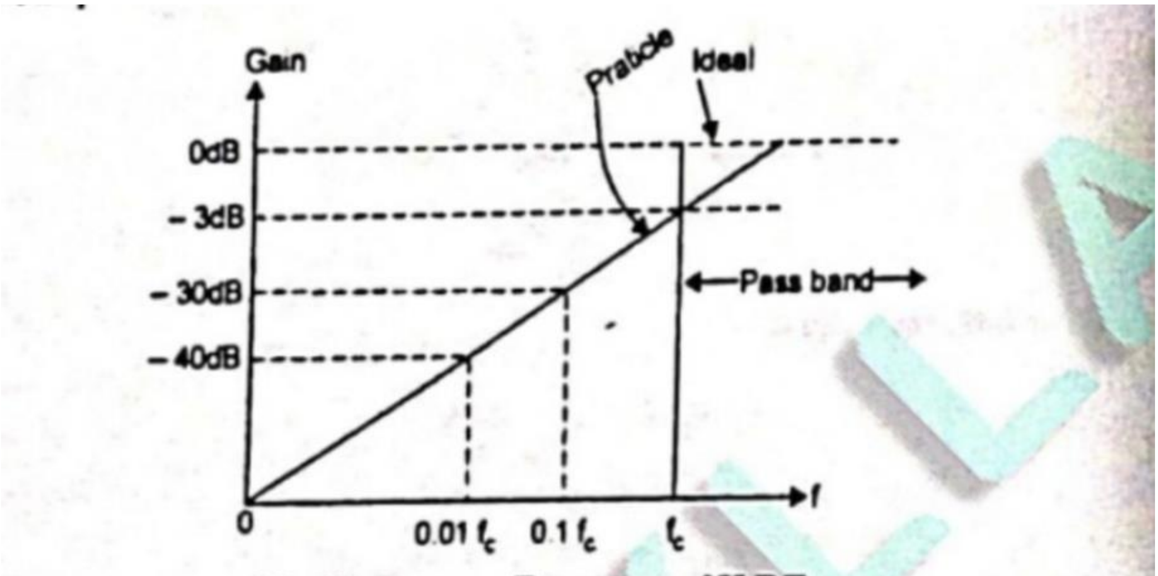


Fig. Frequency Response of H.P.F

Qn10) Discuss wide band pass filter along with its frequency response curve. Also, discuss its frequency.

Ans) Wide Band Pass Filter: A wide band pass filter can be formed by simply cascading high pass and low pass sections and is generally the choice for simplicity of designs and performance though such a circuit can be realized by a number of possible circuits. To form a ± 20 dB/decade band pass filter, a first order high pass and a first order low pass sections are cascaded for a ± 40dB/decade band pass filter, second order high pass filter and a second order low pass filter are connected in series, and so on. It means that the order of the band pass filter is governed by the order of the high pass and low pass filter it consists of.

A ± 20 dB/decade wide band pass filter composed of a first order high pass filter and a first order low pass filter.

|Hhp|=|(1+Rf/Ri) *(j*2pi*f*R2*C2/1+j*2pi*f*R3*C3)